MSc – Data Science and AI

CST4245

Data Visualisation, Computer Vision and Imaging

ASSIGNMENT

"Visualising the Triple Threat: A Dashboard-Driven Study of Worldwide Chronic Disease Prevalence"

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“Visualising the Triple Threat: A Dashboard-Driven Study of Worldwide Chronic Disease Prevalence”

1. Introduction

Data Visualisation plays a crucial role in understanding the complex datasets and communicating the insights accurately and effectively.

In Modern healthcare research, large datasets are collected on a global level to monitor disease prevalence and identification of potential health risks. However, the raw data given was alone difficult to interpret without appropriate visual analysis.

This Project focuses on analysing the global health dataset containing the information related to obesity, diabetes, hypertension, and other non-communicable disease risk factors across different countries, years, and genders.

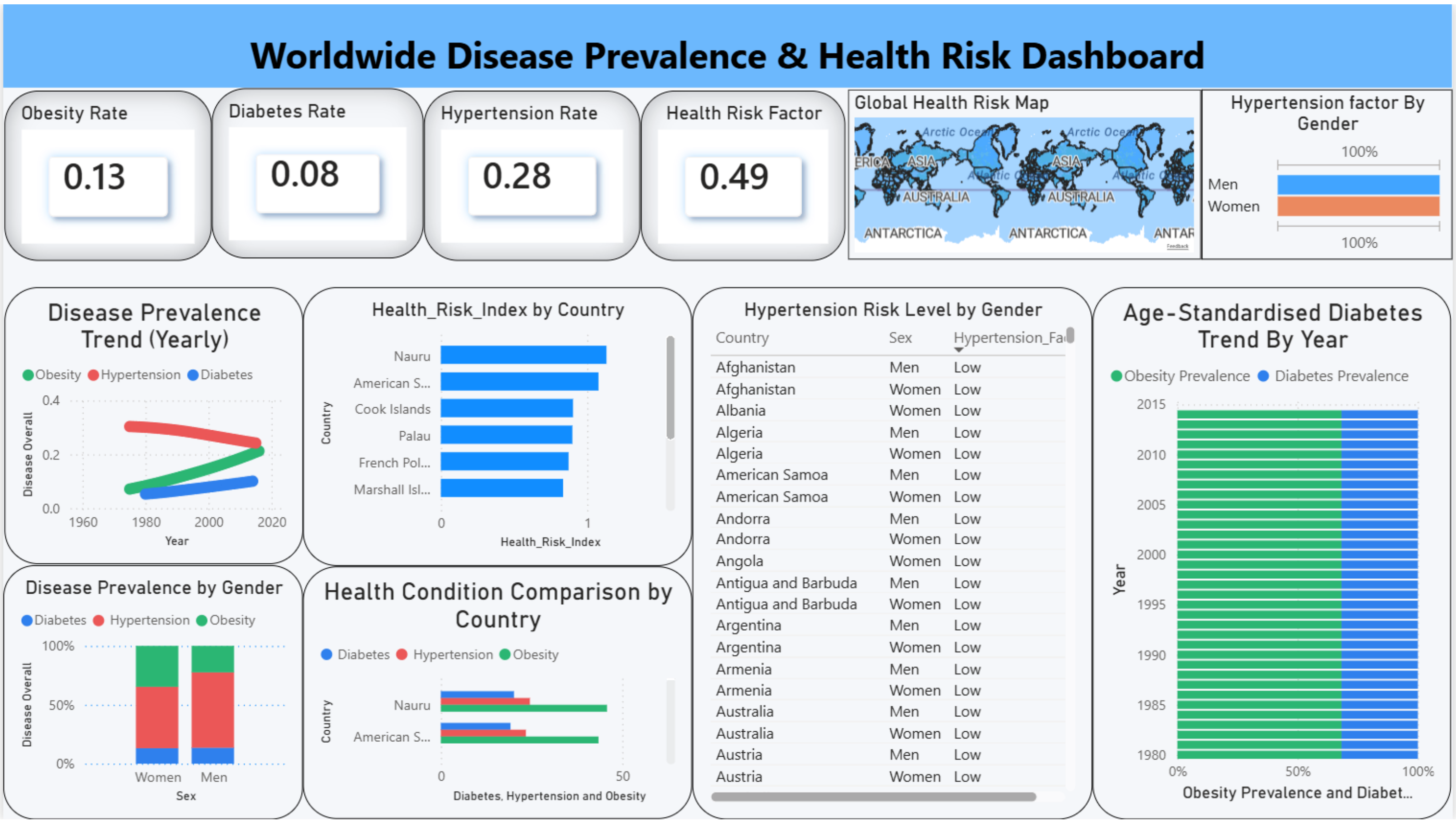
The key reason of this project is to transform raw health data into meaningful insights using Power BI dashboards and visual analytics techniques.

The rapid growth of global health data has created new opportunities for analysing various disease trends and successfully identifies risk factors which affects populations worldwide. Governments, healthcare organisations, and researchers increasingly rely on data analytics to support evidence – based decision making and improving public health outcomes.

Main aims of these projects are:

- To clean and prepare the dataset given for further analysis
- To create a structured data model suitable for analysis
- To design meaningful visualizations using Power Bi
- To identify patterns and insights in global disease prevalence

- To create new measures for better visualisation
- To apply visual encoding principles for effective communication



The developed dashboard titles “Worldwide Disease Prevalence & Health Risk Dashboard” provides an interactive interface for analysing global disease trends. The dashboard enables users to explore patterns in health risks across countries, identifying the differences between genders, and observe long term disease trends.

The project follows the principles of visual encoding, data storytelling, and dashboard design to ensure the visualizations are clear, informative and easy to interpret for both technical and non-technical audiences.

The results of this analysis provide insights into how health risks vary globally and highlight potential areas where healthcare interventions may be required.

2. Dataset Description (The "What")

2.1Dataset Type

The dataset used in this project is a structured dataset consisting of multiple datasets consisting of multiple tables which includes numerical as well as categorical health data.

Structured datasets are organised in rows and columns, making them appropriate for relational modelling and data visualisation tools such as Power Bi or Altair.

The dataset contains information related to:

- Country
- Year
- Gender
- Obesity Prevalence
- Diabetes Prevalence
- Hypertension Prevalence
- Health Risk Indicators

This dataset allows the analysis of temporal, geographic, and demographic health trends.

2.2Data Types

The dataset includes several different types of data:

Numerical Data – Quantitative values representing disease prevalence like Obesity Rate, Diabetes Rate, Hypertension Rate, and Health Risk index.

These values are continuous variables and are essential for statistical comparison and trend analysis.

Categorical Data - data that represents classification variables in this dataset which includes Country, Gender, Risk Level.

These variables are used for grouping and comparison within visualisations.

Temporal Data -This dataset also includes time-based data like Year.

Temporal data allows analysis of disease trends over time.

2.3Attribute Types

Attribute	Type	Description
Country	Nominal	Name of the country
Sex	Nominal	Male / Female
Year	Temporal	Time dimension
Obesity Rate	Continuous	Percentage of obesity prevalence
Diabetes Rate	Continuous	Diabetes prevalence
Hypertension Rate	Continuous	Blood pressure risk
Health Risk Index	Continuous	Combined health risk metric

3. Data preparation

Before creating a dashboard, the raw dataset needs to be prepared in several steps.

3.1Data Cleaning

Real-time datasets often contain inconsistencies that needs to be addressed before meaningful analysis can be performed.

The following cleaning operations were carried out for further analysis.

1. Duplicate records were removed to avoid inflated values in calculations.
2. Missing values were handled by using remove rows options in Power Bi.
3. Country names were standardised across tables to ensure proper relationships in the data model. For example, “United States of America” → “USA”

3.2 Data Transformation

The key transformation required by the dataset to support the analytical queries included:

1. Converting numerical values into consistent percentage formats
2. Generating calculated measure like “High Risk Index” and “Health Risk Factor”
3. Merging the required columns in to a new query known as “Global Metabolic Health”

3.3 Data Integration

Multiple datasets were integrated to create a unified analytical model.

The integrated dataset includes:

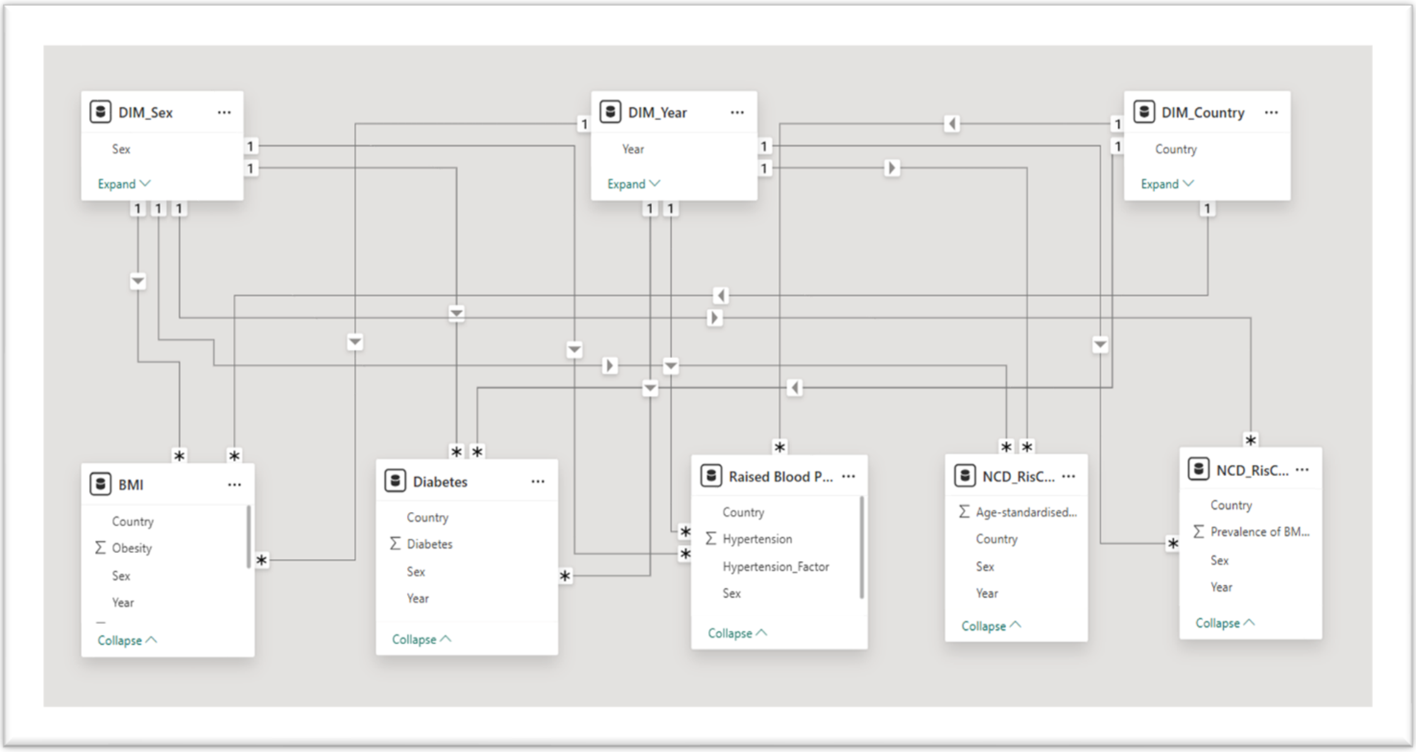
1. BMI (Obesity Prevalence)
2. Diabetes Prevalence
3. Raised blood pressure statistics
4. Non-communicable disease risk indicators

These datasets were linked through common attributes:

- Country
- Year
- Sex

5. Data Model

The Power BI data model follows a star schema structure, which is commonly used in



analytical systems to improve query performance and simplify relationships.

The model consists of three-dimension tables:

- DIM_Country
- DIM_Year
- DIM_Sex

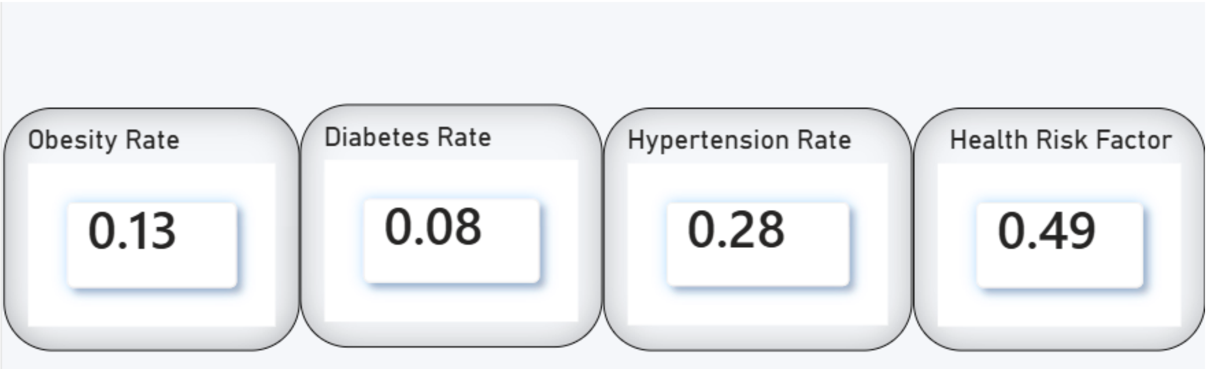
  These dimension tables provide descriptive attributes used for filtering and grouping the data across visualisations.

  The model also contains several fact tables that store numerical metrics:

- BMI table containing obesity prevalence values
- Diabetes table containing diabetes prevalence data
- Raised Blood Pressure table containing hypertension statistics
- NCD Risk table containing additional health risk indicators

Relationship was created between dimension and fact tables using common keys such as Country, Year, and Sex. This structure allows Power Bi to efficiently filter and aggregate data across multiple visualisations.

6. Dashboard Visual Explanations



KPI Cards

These cards provide quick summary indicators:

- Obesity Rate
- Diabetes Rate
- Hypertension Rate
- Health Risk Factor

Purpose: These cards allow users to quickly understand the overall global averaged before exploring detailed data.
Visual Encoding: mark – text; channel: size and position

Global Health Risk Map

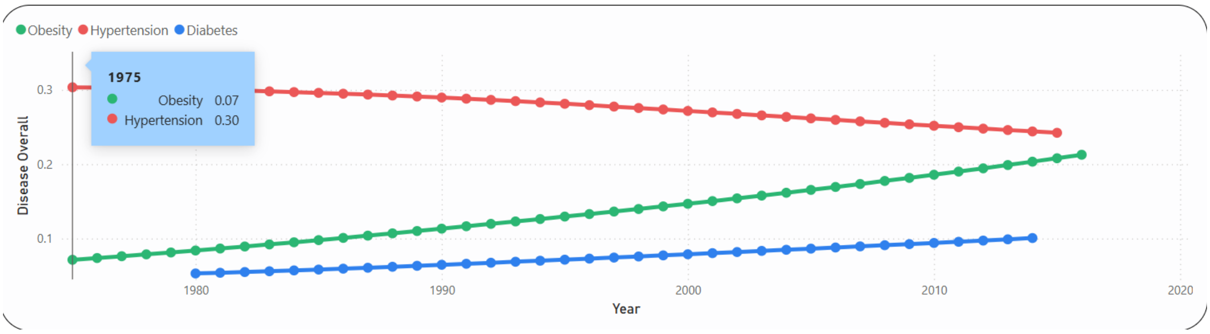
Purpose: This map visualisation displays the geographic distribution of health risks across different countries.

Insight: Users can immediately identify regions with higher disease prevalence.

Visual encoding: Mark- geographic regions; channel: colour intensity

This allows quick spatial comparison of disease prevalence

Disease Prevalence Trend - Line chart

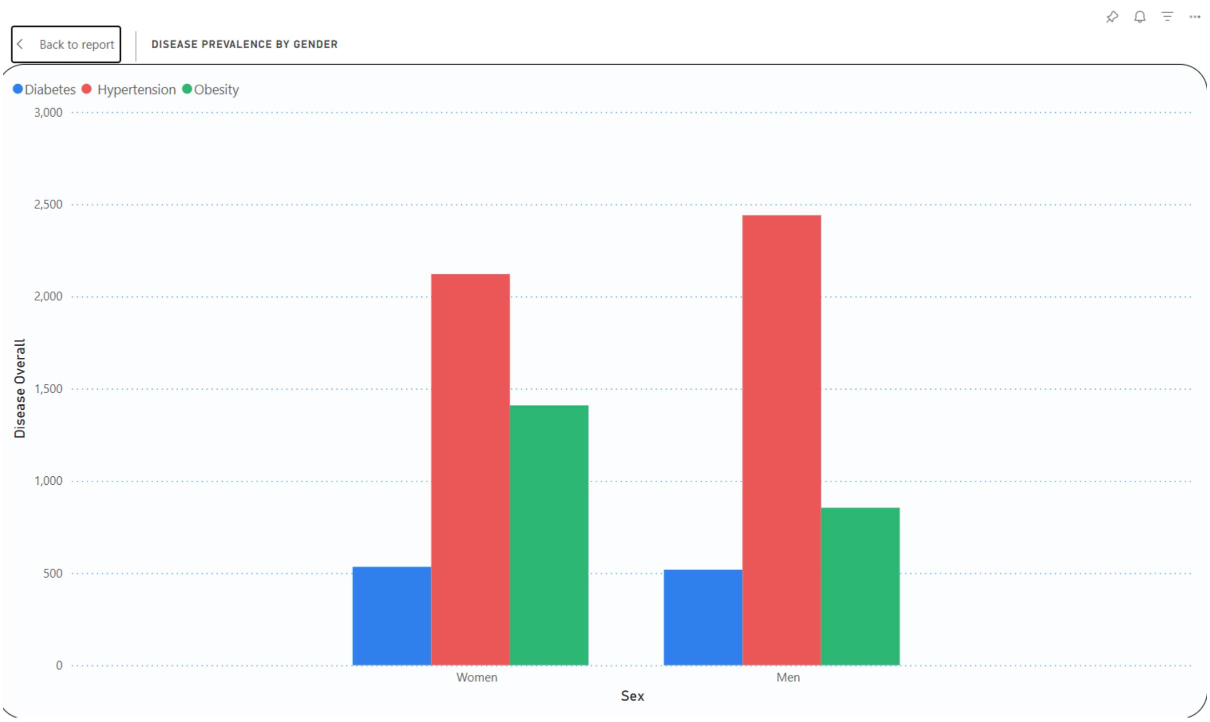


Purpose: To analyse how obesity, diabetes and hypertension have changed with time

Insight: The line chart reveals increasing trends in obesity prevalence globally

Visual Encoding: mark- lines; channel – position on X (year) and Y (prevalence)

Disease Prevalence Trend – Clustered Column Chart



Purpose: The purpose of this visualisation is to compare the distribution of major non-communicable diseases between men and women.

Insight: This column bar chart helps answer key analytical questions:

- Do men or women experience higher levels of certain diseases?
- Which disease has the highest overall prevalence?
- Are there significant gender disparities in health conditions?

By presenting the data side-by-side, the visual allows easy comparison between genders and disease types.

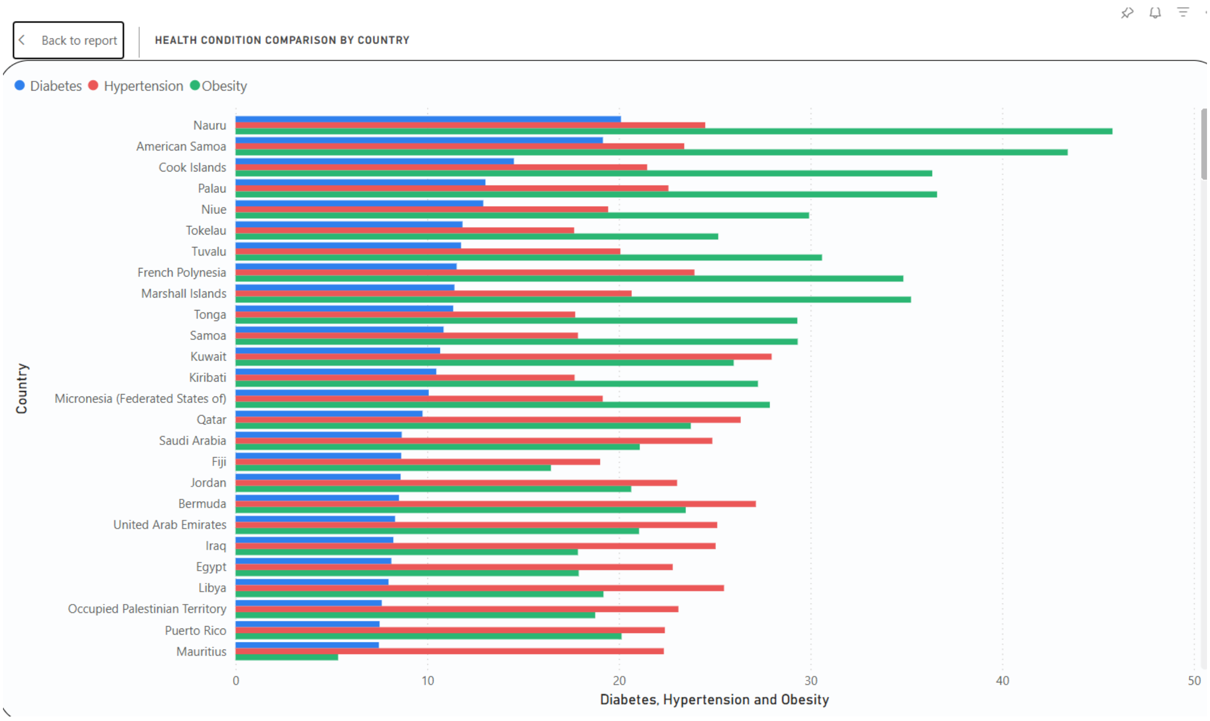
Visual Encoding –

Mark – Bars;

Channel –

- Position (X-axis): Gender categories (Men, Women)
- Position (Y-axis): Disease prevalence values
- Colour: Differentiates Diabetes, Hypertension, and Obesity
- Length (Bar height): Shows magnitude of disease prevalence

Health Condition Comparison by Country – Clustered Bar Chart



Purpose: This chart compares the prevalence of Diabetes, Hypertension, and Obesity across different countries to identify which countries have higher health risk levels**.**

Insight: The chart shows that obesity levels are generally higher than diabetes and hypertension in many countries. This indicates that obesity is a major health risk in these regions.

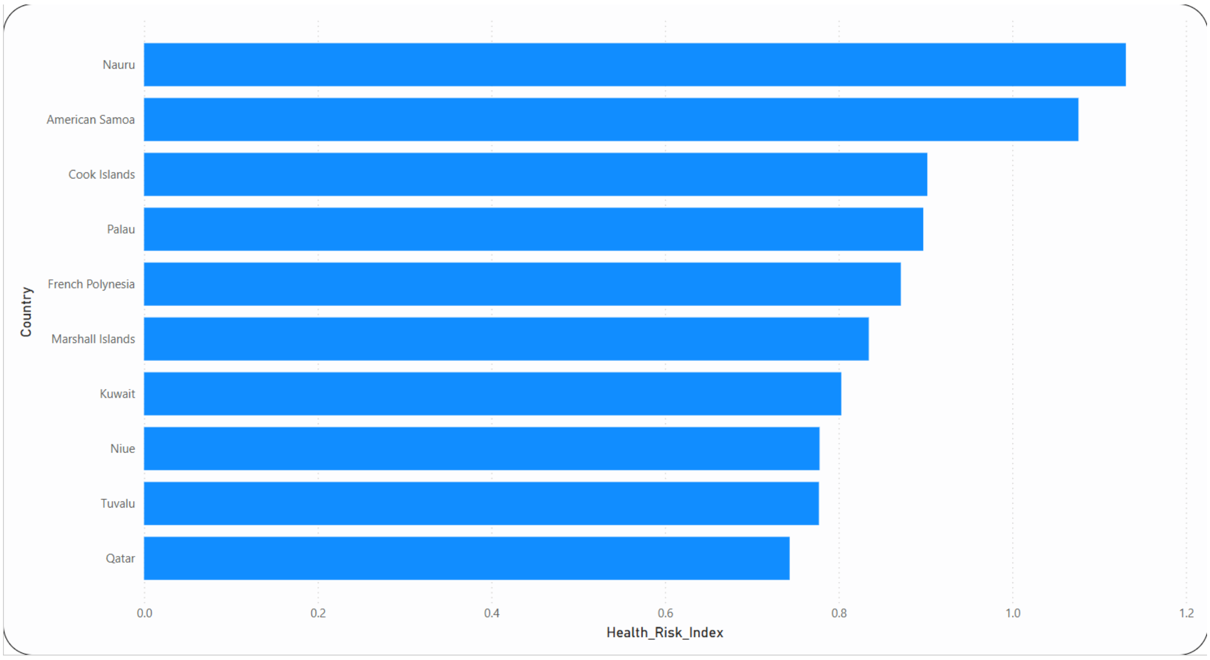
Visual Encoding:

Mark – Bar;

Channels-

- Position (Y-axis): Countries
- Position (X-axis): Disease prevalence values
- Colour: Differentiates Diabetes (blue), Hypertension (red), and Obesity (green)
- Length (Bar size): Shows the magnitude of prevalence

Health Risk Index by Country - Horizontal bar chart



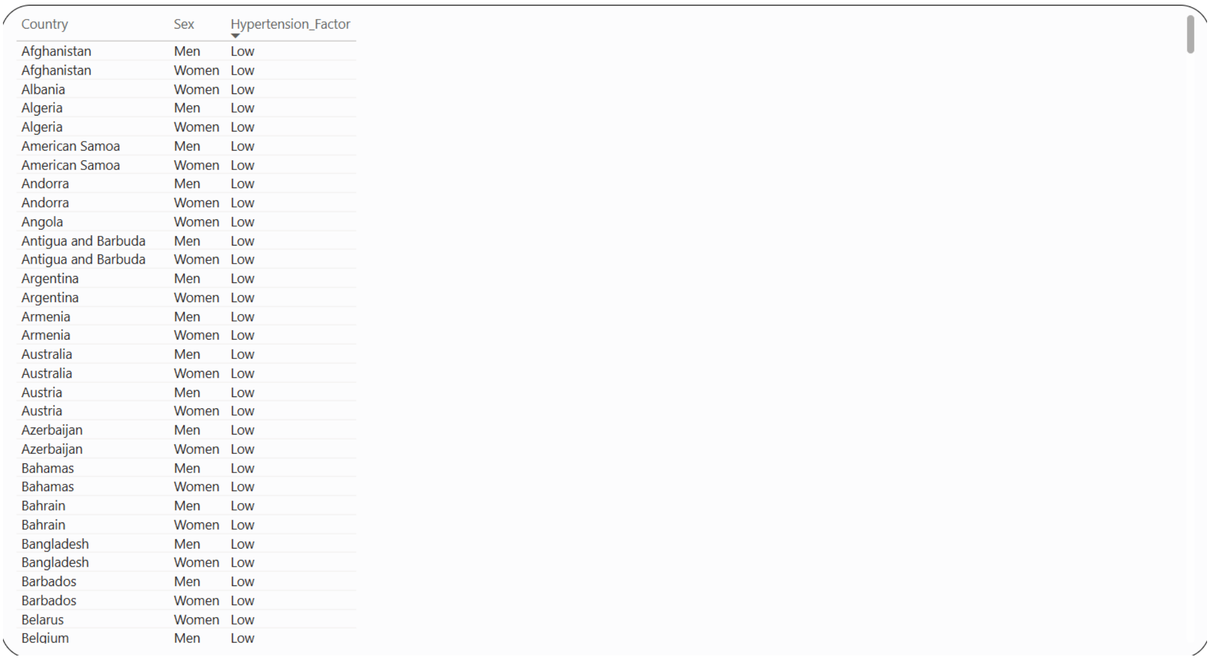
Purpose: To compare the health risks level across several countries (TOP 10)

Insight: Some countries show significantly higher risk levels.

Visual Encoding: Mark- bars; channel – length (risk index)

Bar charts are ideal for category comparisons like Country vs Health Risk Index

Hypertension Risk by Gender - Stacked bar chart



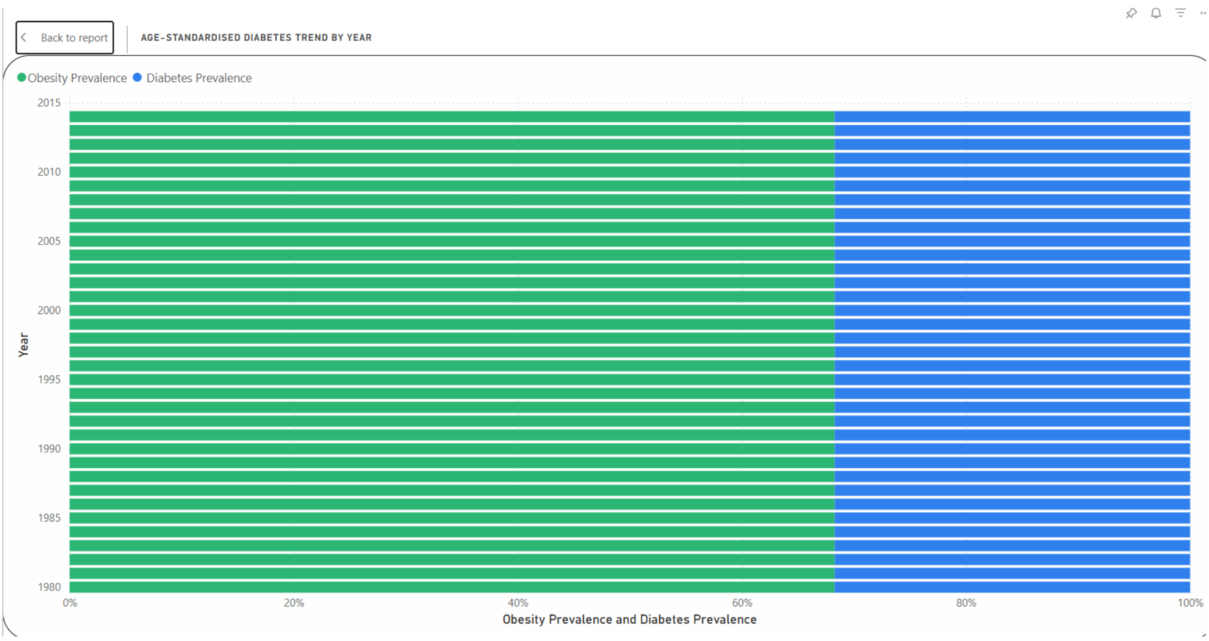
Country	Sex	Hypertension_Factor
Afghanistan	Men	Low
Afghanistan	Women	Low
Albania	Women	Low
Algeria	Men	Low
Algeria	Women	Low
American Samoa	Men	Low
American Samoa	Women	Low
Andorra	Men	Low
Andorra	Women	Low
Angola	Women	Low
Antigua and Barbuda	Men	Low
Antigua and Barbuda	Women	Low
Argentina	Men	Low
Argentina	Women	Low
Armenia	Men	Low
Armenia	Women	Low
Australia	Men	Low
Australia	Women	Low
Austria	Men	Low
Austria	Women	Low
Azerbaijan	Men	Low
Azerbaijan	Women	Low
Bahamas	Men	Low
Bahamas	Women	Low
Bahrain	Men	Low
Bahrain	Women	Low
Bangladesh	Men	Low
Bangladesh	Women	Low
Barbados	Men	Low
Barbados	Women	Low
Belarus	Women	Low
Belgium	Men	Low

Purpose: To compare health risks between men and women.

Insight: The chart highlights the gender differences in diseases prevalence and how gender affects the Hypertension factor.

Visual Encoding: mark – bars; channel – gender

Diabetes and Obesity Trend (Age Standardised) - Area chart



Purpose: To analyse long-term diabetes prevalence trends while accounting for demographic differences.

Insight: The visual describes gradual increases in diabetes prevalence.

Visual Encoding:

Mark – Bar

Channel –

- Position (Y-axis): Year (shows the timeline)
- Position (X-axis): Percentage prevalence values
- Colour: Differentiates Obesity (green) and Diabetes (blue)
- Length (Bar size): Represents the magnitude of prevalence for each disease

7. Calculated measures included in the report

Measure Name	DAX Formula	Purpose
Average Obesity Rate	Avg Obesity Rate = AVERAGE(BMI[Obesity])	Calculates the average prevalence of **Obesity** across the dataset to understand overall obesity levels.
Average Diabetes Rate	Avg Diabetes Rate = AVERAGE(Diabetes[Diabetes])	Computes the mean prevalence of **Diabetes** for comparative health analysis.
Average Hypertension Rate	Avg Hypertension Rate = AVERAGE('Raised Blood Pressure'[Hypertension])	Determines the average prevalence of **Hypertension** across countries and years.
Health Risk Index	Health Risk Index = AVERAGE(BMI[Obesity]) + AVERAGE(Diabetes[Diabetes]) + AVERAGE('Raised Blood Pressure'[Hypertension])	Creates a combined health indicator by aggregating obesity, diabetes, and hypertension prevalence.
Obesity Growth Over Time	Obesity Trend = CALCULATE(AVERAGE(BMI[Obesity]), ALLEXCEPT(BMI, BMI[Year]))	Measures the yearly trend of obesity prevalence to analyse long-term changes over time.

8. Key Insights from the Report

Insight 1 – Hypertension as a Major Risk

In the report Hypertension appears consistently higher than other non- communicable health conditions in several countries, which highlights the risk of a global cardiovascular factor.

Insight 2 – Regional Differences

Certain countries show much more higher health risk index than others. These differences many be influenced by healthcare infrastructure,economic development,and lifestyle patterns.

Insight 3 – Obesity Rates are higher in Pacific Island Countries

The country comparison chart shows that several Pacific island countries such as Nauru and American Samao portray higher levels of Obesity compared other countries in the dataset. This indicates that obesity may be influenced by regional lifestyle factors, diet patterns, and limited access to health interventions in these areas.

Insight 4 – Long Term Trends show increasing Health Risks

According to the time-based analysis both obesity and diabetes prevalence have gradually increased over the years. This trend suggests that no-communicable diseases are becoming more common globally, like due to factors such as urbanisation, lifestyles and diet.

Insight 5 – Diabetes Is relatively stable

Comparing to other conditions, Diabetes shows relatively smaller variation across genders and countries. This suggests that diabetes prevalence is widespread across populations and may be influenced by global lifestyle changes such as diet and reduced physical activity.

9. Action and Targets (The "WHY")

- The insights from the dashboard highlight several health risks related to Obesity, Hypertension, and Diabetes across multiple countries and demographic groups.
- Based on these findings, several actions can be recommended to improve public health outcomes.
- The most affective approach can be promoting healthier diets to reduce obesity and related health conditions.
- Governments and health organisations can encourage balanced nutrition through awareness campaigns and educational programs.
- Increasing public awareness of physical activity is crucial, as sedentary lifestyles are strongly linked to chronic diseases.
- The primary stakeholders who can benefit from these insights include government health departments, healthcare policy makers, and international organisations such as World Health Organization which work to develop strategies for improving global health.

10.Limitations

Although the dashboard provides insights which are useful into global disease prevalence, several limitations should be considered while interpreting the results:

- Some countries contain incomplete records for certain years, which may affect the accuracy of trend analysis. Missing or inconsistent data can lead to potential bias in the findings.
- Secondly, the dataset mainly focuses on the disease prevalence and does not include lifestyle patterns, diet, physical activities or healthcare accessibility which can significantly influence conditions such as Obesity, Diabetes and Hypertension.
- Additionally, the analysis is based on aggregated country level data, which may hide the variations within specific populations. It is very important to note that the analysis identifies corelations between variables rather than cause-and-effect relationships.
- Further studies could improve the analysis by integrating additional datasets containing demographic, and economic indicators.

11.Conclusion

This project demonstrates how data visualisation techniques can be used to transform complex health data into meaningful insights.

Using Microsoft Power BI, an interactive dashboard was developed to analyse worldwide disease prevalence and health risk indicators. The dashboard includes multiple visualisations, including bar charts, maps, and trend charts to present the information in a clear and accessible manner. Through these effective visual encoding techniques such as colour, position, and bar length the dashboards help users to easily compare health conditions across countries, genders, and time periods. The analysis highlights several important patterns, including high levels of Hypertension, increasing trends in Obesity, and relatively stable levels of Diabetes across populations. Overall, the project shoes how interactive dashboards can support data-driven decision making and help policy makers design the new strategies **to improve the global health outcomes.**